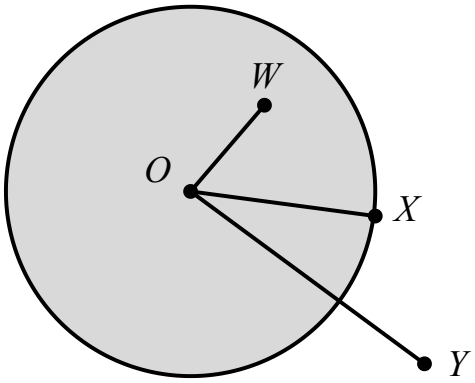
STPM 2021 PHYSICS 960/2

Questions and Answers

Values of constants: $\varepsilon_0 = 8.85 \times 10^{-12}\ F\ m^{-1}$ ($\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\ N\ m^2\ C^{-2}$), $\mu_0 = 4\pi \times 10^{-7}\ H\ m^{-1}$.

Section A [15 marks]

1. A charged metal sphere with its centre, *O*, is shown in the diagram below.



If the electric field strength at points *W*, *X* and *Y* are E_W, E_X and E_Y respectively, which is the correct arrangement of E_W, E_X and E_Y?

A $E_W < E_X < E_Y$

B $E_W > E_X > E_Y$

C $E_X < E_Y < E_W$

D $E_X > E_Y > E_W$

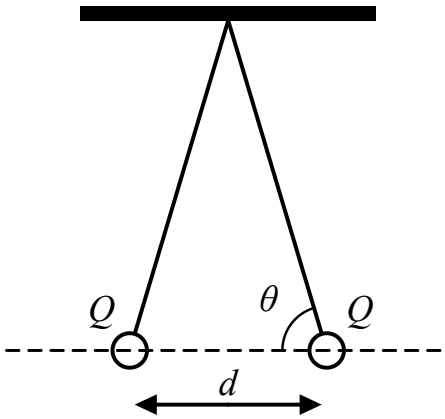
Answer: D

Inside a conductor $E = 0$ (*W*); at the surface it is largest (*X*); outside it falls as $1/r^2$ (*Y*).

$$E_X > E_Y > E_W$$

Explanation: The charge on a conductor sits on its surface.

2. A simple experimental setup which consists of two small hollow metal spheres can be used to verify Coulomb's law. An equal quantity of charges is supplied to the metal spheres and the metal spheres are suspended by two nylon strings as shown in the diagram below.



If the strings incline at an angle, θ, with the horizontal, the charges and mass on each sphere is Q and m respectively, what is the distance, d?

A $\sqrt{\left(\frac{Q^2 \sin\theta}{4\pi\varepsilon_o mg}\right)}$

B $\sqrt{\left(\frac{4\pi\varepsilon_o mg}{Q^2 \sin\theta}\right)}$

C $\sqrt{\left(\frac{Q^2 \tan\theta}{4\pi\varepsilon_o mg}\right)}$

D $\sqrt{\left(\frac{4\pi\varepsilon_o mg}{Q^2 \tan\theta}\right)}$

Answer: C

$$\text{Vertical: } T\sin\theta = mg;\ \text{horizontal: } T\cos\theta = \frac{Q^2}{4\pi\varepsilon_o d^2}$$

$$\frac{Q^2}{4\pi\varepsilon_o d^2} = \frac{mg}{\tan\theta} \Rightarrow d = \sqrt{\left(\frac{Q^2 \tan\theta}{4\pi\varepsilon_o mg}\right)}$$

Explanation: θ is measured from the horizontal, so the vertical component of tension is $T\sin\theta$.

3. A parallel plate capacitor of capacitance C is connected to a battery of e.m.f. V. The capacitor is charged to charge Q. A piece of insulator is then inserted into the capacitor while the battery is still connected. Which statement is true about the capacitor after the insertion of the insulator?

A The charges in the capacitor is greater than Q.
B The capacitance of the capacitor is less than C.
C The total energy of the capacitor remains constant.
D The potential difference across the capacitor is less than V.

Answer: A

With the battery connected V is fixed; the insulator (dielectric) increases C, so $Q = CV$ becomes greater than before.

Explanation: The energy $\frac{1}{2}CV^2$ also increases.

4. A flash-lamp of a camera gains energy from a capacitor of 150 μF which has been charged to 1000 V. If all the energy stored is converted to light and the light is emitted for 0.040 s, what is the average power of the light?

A 3.0 W B 75 W C 1.9×10^3 W D 3.8×10^3 W

Answer: C

$$U = \frac{1}{2}CV^2 = \frac{1}{2}(150 \times 10^{-6})(1000)^2 = 75 \text{ J}$$

$$P = \frac{75}{0.040} = 1.9 \times 10^3 \text{ W}$$

Explanation: Average power = energy ÷ time.

5. When an electric field is applied across a metal conductor, the electrical conduction in the conductor is the result of

A the random movement of free electrons at high speed
B the collision between free electrons and ions
C the net velocity of free electrons
D the vigorous vibration of ions

Answer: C

The current is due to the net (drift) velocity of the free electrons along the conductor.

Explanation: Random motion alone gives no net flow of charge.

6. Which statement is true about current density for a wire conductor?

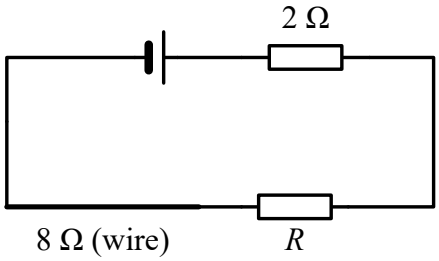
A The current density increases when the electric field of the wire is reduced.
B The current density of the conductor does not depend on the radius of its wire.
C The current density occurs when electrons traverse the cross-sectional area of the wire.
D The current density increases when a steady current flows through a larger cross-sectional area of the wire.

Answer: C

Current density is the current per unit cross-sectional area: it describes charge (electrons) passing through the cross-section of the wire.

Explanation: $J = I/A$: for a steady current a larger area means a smaller current density.

7. A potentiometer consists of a cell of e.m.f. 2.00 V with an internal resistance of 2.00 Ω, a slide wire of resistance of 8.00 Ω and a resistor connected in series. What is the required resistance of the resistor to produce a potential difference of 50.0 mV across the wire?



A 308 Ω B 310 Ω C 312 Ω D 320 Ω

Answer: B

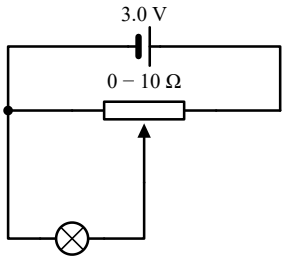
$$\text{Current needed: } I = \frac{50.0 \times 10^{-3}}{8.00} = 6.25 \times 10^{-3}\ \text{A}$$

$$\text{Total resistance} = \frac{2.00}{6.25 \times 10^{-3}} = 320\ \Omega \Longrightarrow R = 320 - 2.00 - 8.00 = 310\ \Omega$$

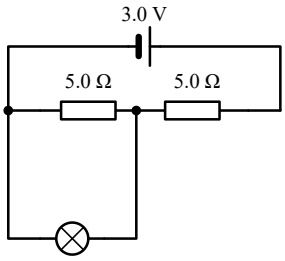
Explanation: A large series resistor gives a small p.d. across the wire, for measuring small e.m.f.s.

8. Which circuit diagram is suitable to adjust the brightness of the bulb labelled 3.0 V, 5.0 W from zero to normal brightness?

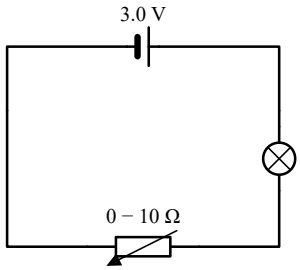
A



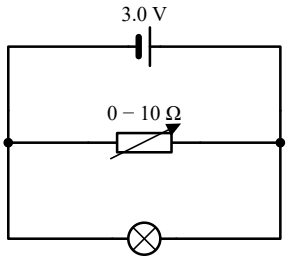
B



C



D



Answer: A

A potential divider (the rheostat used with three terminals, circuit A) can give the bulb any p.d. from 0 V up to the full 3.0 V.

Explanation: A series rheostat cannot reduce the current to zero.

9. Which statement explains the existence of magnetic force between two current-carrying conductors parallel to each other?

A The resultant magnetic field produced by the currents in both conductors acts on each conductor.
B The magnetic field produced by the current in one conductor acts on the other conductor.
C The magnetic field produced by the current in each conductor acts on itself.
D An external magnetic field acts on both conductors.

Answer: B

Each current produces a magnetic field at the position of the other conductor, which exerts a force on it.

Explanation: A conductor does not experience a force from its own field.

10. A charged particle enters a region consisting of crossed magnetic and electric fields. The magnetic flux density is 0.092 T and the electric field strength is 3.67×10^5 V m^{-1}. If the charged particle is not deflected, what is the velocity of the particle?

A 3.0×10^8 m s^{-1} B 4.0×10^6 m s^{-1} C 3.4×10^4 m s^{-1} D 2.5×10^{-7} m s^{-1}

Answer: B

$$qE = qvB \Longrightarrow v = \frac{E}{B} = \frac{3.67 \times 10^5}{0.092} = 4.0 \times 10^6 \text{ m s}^{-1}$$

Explanation: The velocity selector does not depend on the charge or mass.

11. Which is **not** true about the application of Hall effects?

A Compare the e.m.f.s of two dry cells
B Determine the density of charge carriers in a conductor
C Compare the magnetic flux density of two magnetic fields
D Determine the majority charge carriers in a semiconductor

Answer: A

The Hall effect cannot be used to compare the e.m.f.s of cells (a potentiometer does that). Statement A is not true.

Explanation: Hall effect uses: measuring B, carrier density and carrier type.

12. A plane coil of N turns is placed in a uniform magnetic field of magnetic flux density B. The normal of the surface of the coil makes an angle θ with the direction of the magnetic field. If the area of the coil is A, what is the magnetic flux through the coil?

A BA B $BA \cos \theta$ C NBA D $NBA \cos \theta$

Answer: D

Total flux through the N turns (flux linkage) = $NBA\cos\theta$

Explanation: θ is measured from the normal, so the cosine is used. The flux through one turn is $BA\cos\theta$; the expected answer counts all N turns.

13. A back e.m.f. of 0.85 V is induced when a current in an inductor increases from zero to 3.5 A in 6.0 ms. If the inductor carries the current of 3.5 A, what is the energy stored?

A 17.8 mJ B 8.9 mJ C 5.1 mJ D 2.6 mJ

Answer: B

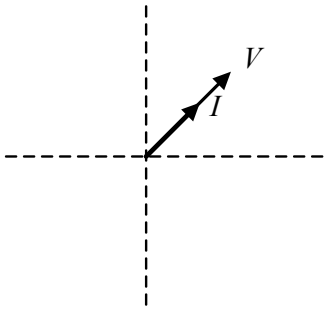
$$L = \frac{\varepsilon}{\Delta I/\Delta t} = \frac{0.85}{3.5/6.0 \times 10^{-3}} = 1.46 \times 10^{-3}\ \text{H}$$

$$U = \frac{1}{2}LI^2 = \frac{1}{2}(1.46 \times 10^{-3})(3.5)^2 = 8.9\ \text{mJ}$$

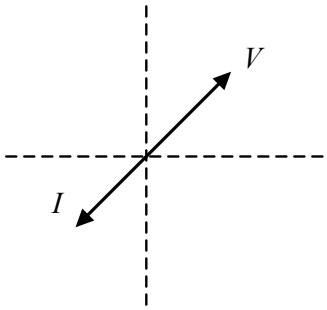
Explanation: Find L from the back e.m.f., then use the energy formula.

14. Which is the correct phasor diagram for the voltage, V, and current, I, if the resistor is connected to an a.c. supply?

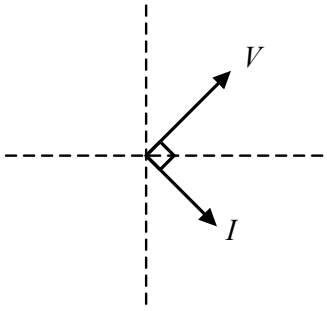
A



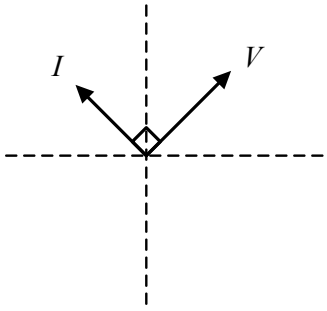
B



C



D



Answer: A

In a resistor the current and voltage are in phase, so their phasors point in the same direction (A).

Explanation: Phasors of different lengths can still be in phase.

15. A light bulb is connected to a sinusoidal voltage supply of peak voltage 100 V and the average power dissipated is 50 W. What is the average power dissipated if the bulb is connected to a r.m.s. voltage supply of 80 V?

A 32 W B 50 W C 64 W D 128 W

Answer: C

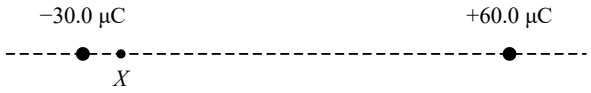
$$R = \frac{V_0^2}{2P} = \frac{100^2}{2(50)} = 100\ \Omega$$

$$P = \frac{V_{rms}^2}{R} = \frac{80^2}{100} = 64\ \text{W}$$

Explanation: Mean power uses r.m.s. values; 100 V peak is 70.7 V r.m.s.

Section B [15 marks]

16. Two point charges, −30.0 μC and +60.0 μC, are separated by a distance of 14.0 cm as shown in the diagram below.



X is located 2.0 cm from the negative charge.

(a) Calculate the force between the charges. [2]

Answer

$F = \frac{q_1 q_2}{4\pi\varepsilon_0 r^2} = \frac{(8.99 \times 10^9)(30.0 \times 10^{-6})(60.0 \times 10^{-6})}{(0.140)^2}$ 1

$F = 826$ N (attractive) 1

2

Explanation: Unlike charges attract.

(b) Determine the magnitude and direction of the resultant electric field at X. [4]

Answer

From −30.0 μC (0.020 m): $E_1 = \frac{(8.99 \times 10^9)(30.0 \times 10^{-6})}{(0.020)^2} = 6.74 \times 10^8$ N C^{-1}, towards the negative charge 1

From +60.0 μC (0.120 m): $E_2 = \frac{(8.99 \times 10^9)(60.0 \times 10^{-6})}{(0.120)^2} = 3.75 \times 10^7$ N C^{-1}, away from the positive charge 1

Both point towards the negative charge: $E = 6.74 \times 10^8 + 0.375 \times 10^8$ 1

$E = 7.12 \times 10^8$ N C^{-1}, directed towards the −30.0 μC charge 1

4

Explanation: Between unlike charges the fields add.

(c) Predict a possible position of a point where the resultant electric field is zero. [1]

Answer

On the line, outside the charges on the side of the smaller (negative) charge: about 33.8 cm beyond the −30.0 μC charge — 1

1

Total for question **7**

Explanation: From $\frac{30}{x^2} = \frac{60}{(x+14)^2}$, $x = 14/(\sqrt{2}-1) = 33.8$ cm.

17. A capacitor of capacitance 10 μF initially is charged by a battery of potential difference 500 V.

(a) Determine the charge in the capacitor. [2]

Answer

$Q = CV = (10 \times 10^{-6})(500)$ — 1

$= 5.0 \times 10^{-3}$ C — 1

2

Explanation: The capacitor is fully charged to the battery p.d.

(b) The battery is then removed and the capacitor is connected to an uncharged capacitor of capacitance 40 μF.

(i) Determine the potential difference across both capacitors. [2]

Answer

Charge is shared, total unchanged: $V = \frac{Q}{C_1 + C_2} = \frac{5.0 \times 10^{-3}}{50 \times 10^{-6}}$ — 1

$V = 100$ V — 1

2

Explanation: The capacitors end up in parallel at the same p.d.

(ii) Calculate the total energy of both capacitors. [2]

Answer

$U = \frac{1}{2}(C_1 + C_2)V^2 = \frac{1}{2}(50 \times 10^{-6})(100)^2$ — 1

$U = 0.25$ J — 1

2

Explanation: Initially 1.25 J was stored; 1.0 J is lost as heat and radiation in the connecting wires.

(iii) The system of the capacitors is then connected to a resistor. Describe the change in the total charge stored in the system with time. [2]

Answer

The capacitors discharge through the resistor	1
The total charge decreases exponentially with time ($Q = Q_0 e^{-t/RC}$) towards zero	1
	2
Total for question	**8**

Explanation: The time constant is R(50 μF).

Section C [30 marks]

18. (a) An electron moves into a uniform magnetic field. Describe the path of the electron when the direction of motion of the electron is

(i) perpendicular to the magnetic field, [4]

Answer

The magnetic force evB is always perpendicular to the velocity	1
so it changes the direction but not the speed	1
The electron moves in a circle	1
of radius $r = \frac{mv}{eB}$ at constant speed	1
	4

Explanation: The force provides the centripetal force.

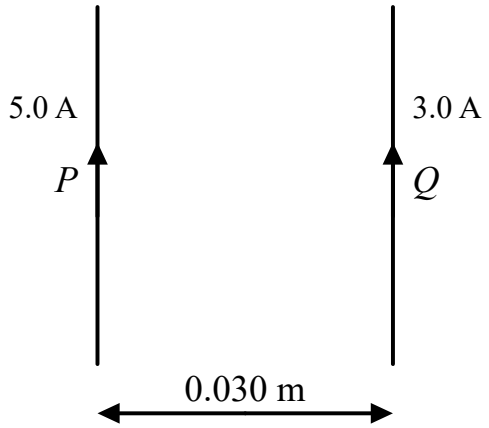
(ii) parallel to the magnetic field. [3]

Answer

There is no magnetic force ($\sin 0° = 0$)	1
so the electron is not deflected	1
It moves in a straight line at constant velocity	1
	3

Explanation: $F = evB \sin \theta$.

(b) Two long parallel wires, P and Q, are separated by a distance of 0.030 m, each wires carries current of 5.0 A and 3.0 A respectively as shown in the diagram below.



(i) Determine the position of a point where the resultant magnetic field is zero. [5]

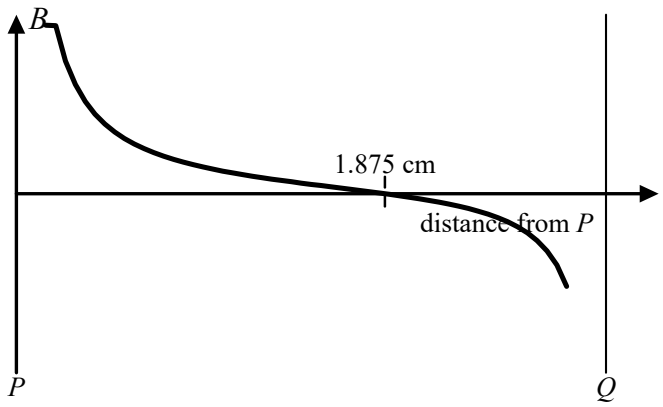
Answer

Both currents are in the same direction, so their fields are opposite between the wires: the zero point lies between them	1
$\frac{\mu_0(5.0)}{2\pi x} = \frac{\mu_0(3.0)}{2\pi(0.030 - x)}$	1
$5.0(0.030 - x) = 3.0x$	1
$x = 0.01875$ m	1
The point is 1.875 cm from P (1.125 cm from Q), between the wires	1
	5

Explanation: It is closer to the wire with the smaller current.

(ii) Sketch a graph of the variation of the resultant magnetic field with the distance between P and Q. [3]

Answer



Large near each wire, opposite directions on either side of the zero point	1
Zero at 1.875 cm from P	1
Correct shape (falls from P, rises towards Q)	1
	3
Total for question	**15**

Explanation: Each wire's field varies as $1/r$.

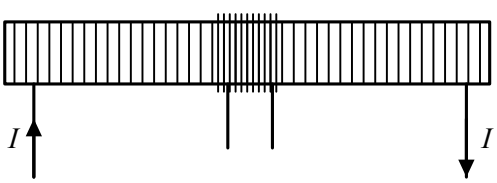
19. (a) Explain the phenomenon of mutual induction. [4]

Answer

Two coils are placed close to each other	1
A changing current in one coil produces a changing magnetic flux	1
which links the second coil	1
inducing an e.m.f. in the second coil: $\varepsilon_2 = -M\,\mathrm{d}I_1/\mathrm{d}t$	1
	4

Explanation: This is the principle of the transformer.

(b) A coil with 10 turns is wound over the middle part of a long solenoid with 1250 turns per meter and diameter 4.0 cm as shown in the diagram below.



The current, I, in the long solenoid is increased at a rate of 100 A s^{-1}.

(i) Determine the magnitude of the induced e.m.f. in the coil. [5]

Answer

Field in the solenoid: $B = \mu_0 nI$ — 1

Flux through the coil: $\Phi = \mu_0 nIA$, $A = \pi(0.020)^2$ — 1

$\varepsilon = N\frac{d\Phi}{dt} = N\mu_0 nA\frac{dI}{dt}$ — 1

$\varepsilon = (10)(4\pi \times 10^{-7})(1250)\pi(0.020)^2(100)$ — 1

$\varepsilon = 1.97 \times 10^{-3}$ V — 1

5

Explanation: The coil encloses the solenoid's full cross-section.

(ii) Determine the mutual inductance. [2]

Answer

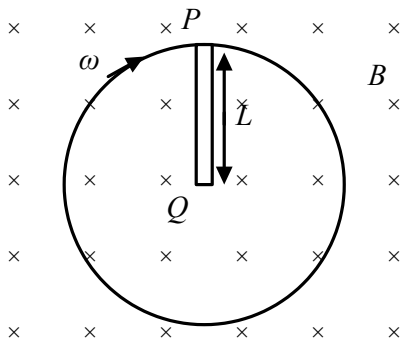
$M = \frac{\varepsilon}{dI/dt} = \frac{1.97 \times 10^{-3}}{100}$ — 1

$M = 1.97 \times 10^{-5}$ H — 1

2

Explanation: Mutual inductance = induced e.m.f. per unit rate of change of current.

(c) A metal rod PQ of length L is spun about its end, Q, at a constant angular speed, ω, in a uniform magnetic field, B, directed into the page as shown in the diagram below.



(i) Derive an expression for the induced e.m.f. between both ends of the rod in terms of ω, L and B. [3]

Answer

In time Δt the rod sweeps an area $\Delta A = \frac{1}{2}L^2\omega\Delta t$ — 1

$\varepsilon = B\frac{\Delta A}{\Delta t}$ — 1

$\varepsilon = \frac{1}{2}B\omega L^2$ — 1

3

Explanation: Equivalently, integrate $Bv\,\mathrm{d}r$ with $v = \omega r$.

(ii) State which end of the rod is at a higher potential. [1]

Answer

For the rotation shown (clockwise with **B** into the page), the force on positive charges is towards the outer end, so P is at the higher potential — 1

1

Total for question **15**

Explanation: Use $\mathbf{F} = q\mathbf{v} \times \mathbf{B}$ for the charges in the moving rod; reversing the rotation would reverse the answer.

20. (a) What is meant by an *alternating current*? [1]

Answer

A current that reverses direction periodically, varying sinusoidally with time — 1

1

Explanation: Its average over one cycle is zero.

(b) A sinusoidal voltage supply, $V = V_o \sin(\omega t)$, is connected to a capacitor of capacitance C.

(i) Derive an expression for the current through the capacitor. [3]

Answer

$Q = CV = CV_o \sin \omega t$ — 1

$I = \frac{\mathrm{d}Q}{\mathrm{d}t} = \omega CV_o \cos \omega t$ — 1

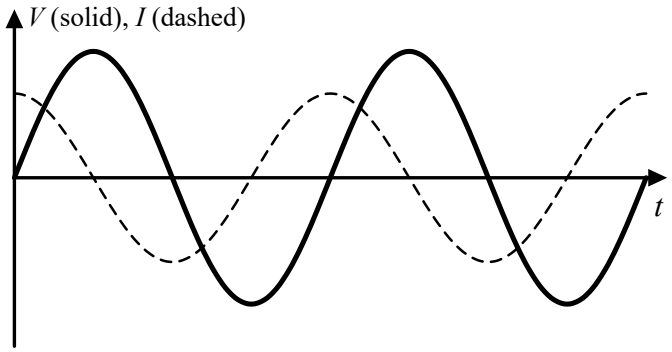
$I = I_o \sin(\omega t + \frac{\pi}{2})$, with $I_o = \omega CV_o$ — 1

3

Explanation: The current leads the voltage by 90°.

(ii) Sketch, on the same axis of time, t, the variation of voltage, V, and the variation of current, I. [2]

Answer



Both sinusoidal with the same period; current leading the voltage by a quarter cycle	2
	2

Explanation: When V is maximum the current is zero.

(iii) Explain the average power in the capacitor. [2]

Answer

The average power is zero	1
Energy stored in the capacitor during one quarter cycle is returned to the supply in the next quarter cycle	1
	2

Explanation: $P = VI = \frac{1}{2}V_oI_o \sin 2\omega t$, which averages to zero.

(c) A sinusoidal voltage, $V = 50 \sin (100t)$, is supplied to a capacitor of capacitance 100 μF and a resistor of resistance 70 Ω connected in series.

(i) Determine the impedance of the circuit. [3]

Answer

$\omega = 100$ rad s^{-1}; $X_C = \frac{1}{\omega C} = \frac{1}{(100)(100 \times 10^{-6})} = 100\ \Omega$	1
$Z = \sqrt{(70^2 + 100^2)}$	1
$Z = 122\ \Omega$	1
	3

Explanation: Read ω directly from the equation.

(ii) Determine the maximum current in the circuit. [2]

Answer

$I_0 = \frac{V_0}{Z} = \frac{50}{122}$ 1

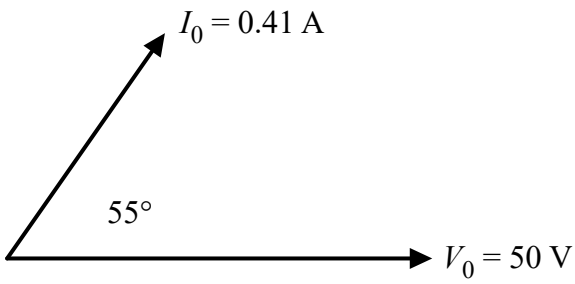
$I_0 = 0.41$ A 1

2

Explanation: The peak voltage is 50 V.

(iii) Draw a phasor diagram for the voltage supply and current in the circuit. [2]

Answer



Current phasor leading the voltage phasor by $\tan^{-1}(100/70) = 55°$ 2

2

Total for question **15**

Explanation: In an *RC* circuit the current leads.